

Noise & Edges and Contours (Part 2)

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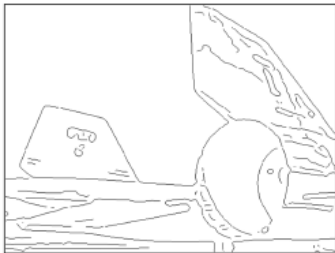
- Noise
- Edges and Contours

Edge?

- What is an Edge: sharp change in brightness (discontinuities)
- Image processing task that finds edges and contours in images
- Edges so important that human vision can reconstruct edge lines



(a)

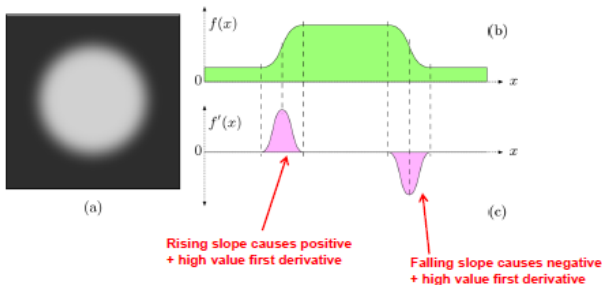


(b)

Characteristics of an Edge

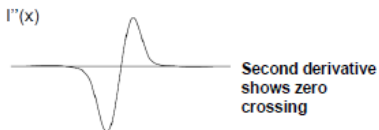
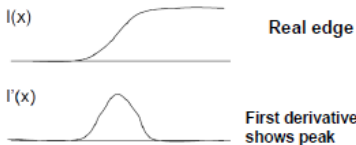
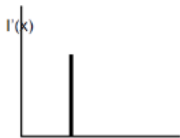
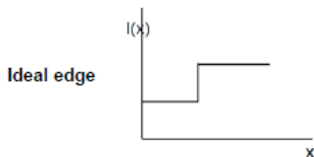
- Edge: A sharp change in brightness
- Ideal edge is a step function in some direction
- Real (non-ideal) edge is a slightly blurred step function
- Edges can be characterized by high value first derivative

$$f'(x) = \frac{df}{dx}(x)$$



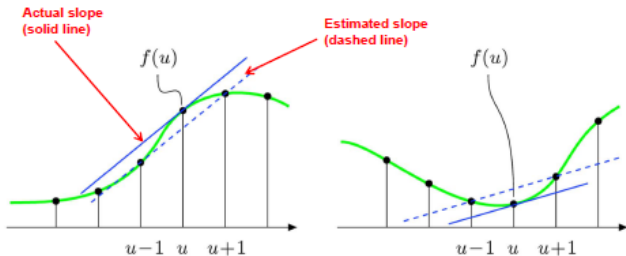
Characteristics of an Edge

- Ideal edge is a step function in certain direction.
- First derivative of $I(x)$ has a **peak** at the edge
- Second derivative of $I(x)$ has a **zero crossing** at edge



Computing Derivative of Discrete Function

$$\frac{df}{du}(u) \approx \frac{f(u+1) - f(u-1)}{2} = 0.5 \times (f(u+1) - f(u-1))$$



- Forward difference (right slope)

$$\Delta_+ f(x) = f(x+1) - f(x)$$

- Backward difference (left slope)

$$\Delta_- f(x) = f(x) - f(x-1)$$

- Central Difference (average slope)

$$\Delta f(x) = \frac{1}{2}(f(x+1) - f(x-1))$$

Definition: Function Gradient

- Let $f(x, y)$ be a 2D function
- Gradient: Vector whose direction is in direction of maximum rate of change of f and whose magnitude is maximum rate of change of f
- Gradient is perpendicular to edge contour

$$\Delta f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right)^T$$

- magnitude = $\left[\left(\frac{\partial f}{\partial x} \right)^2 + \left(\frac{\partial f}{\partial y} \right)^2 \right]^{1/2}$
- direction = $\tan^{-1} \left(\frac{\frac{\partial f}{\partial y}}{\frac{\partial f}{\partial x}} \right)$

Image Gradient

- Image is 2D discrete function

$$\frac{\partial I}{\partial u}(u, v) \text{ and } \frac{\partial I}{\partial v}(u, v)$$

- Image derivatives in horizontal and vertical directions
- Image gradient at location (u, v)

$$\Delta I(u, v) = \begin{pmatrix} \frac{\partial I}{\partial u}(u, v) \\ \frac{\partial I}{\partial v}(u, v) \end{pmatrix}$$

- Gradient magnitude

$$|\Delta I|(u, v) = \sqrt{\left(\frac{\partial I}{\partial u}(u, v)\right)^2 + \left(\frac{\partial I}{\partial v}(u, v)\right)^2}$$

- Magnitude is invariant under image rotation, used in edge detection

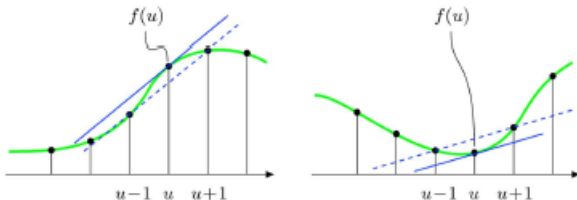
Derivative Filters

- Recall that we can compute derivative of discrete function as

$$\frac{df}{du}(u) \approx \frac{f(u+1) - f(u-1)}{2} = 0.5 \cdot (f(u+1) - f(u-1))$$

- Can we make linear filter that computes central differences

$$H_x^D = [-0.5 \ \mathbf{0} \ 0.5] = 0.5 \cdot [-1 \ \mathbf{0} \ 1]$$



- Forward difference

$$\Delta_+ f(x) = f(x+1) - f(x)$$

- Take a convolution kernel $H = [0 \ -1 \ 1]$

$$\Delta_+ f = f * H$$

- Central Difference (average slope)

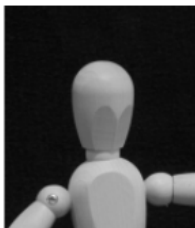
$$\Delta f(x) = \frac{1}{2}(f(x+1) - f(x-1))$$

- Convolution kernel is $H = [-0.5 \ 0 \ 0.5]$

$$\Delta f(x) = f * H$$

- **Notice:** Derivative kernel sum to zero

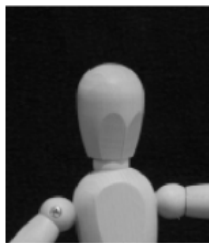
x-Derivative of Image using Central Difference



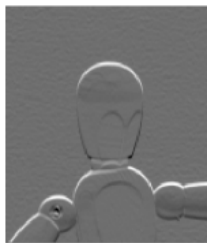
$$\star \begin{bmatrix} -0.5 & 0 & 0.5 \end{bmatrix} =$$



y-Derivative of Image using Central Difference



$$* \begin{bmatrix} -0.5 \\ 0 \\ 0.5 \end{bmatrix} =$$



- Approximating local gradients in image is basis of many classical edge-detection operators
- Main differences?
 - Type of filter used to estimate gradient components
 - How gradient components are combined
- We are typically interested in
 - Local edge direction
 - Local edge magnitude

Partial Image Derivatives

- Partial derivatives of images replaced by finite differences

$$\Delta_x f = f(x, y) - f(x - 1, y) \longrightarrow \begin{pmatrix} -1 & 1 \end{pmatrix}$$

$$\Delta_y f = f(x, y) - f(x, y - 1) \longrightarrow \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

- Alternatives are:

$$\Delta_{2x} f = f(x + 1, y) - f(x - 1, y) \longrightarrow \begin{pmatrix} -1 & 0 & 1 \end{pmatrix}$$

$$\Delta_{2y} f = f(x, y + 1) - f(x, y - 1) \longrightarrow \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$$

- Robert's gradient

$$\Delta_+ f = f(x + 1, y + 1) - f(x, y) \longrightarrow \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

$$\Delta_- f = f(x, y + 1) - f(x + 1, y) \longrightarrow \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

- Prewitt

$$\begin{pmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{pmatrix}$$
$$\begin{pmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix}$$

- Sobel

$$\begin{pmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{pmatrix}$$
$$\begin{pmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{pmatrix}$$

Using Averaging with Derivatives

- Finite difference operator is sensitive to noise
- Derivates more robust if derivative computations are averaged in a neighborhood
- Prewitt operator: derivative in x , then average in y

$$H_x^P = \frac{1}{3} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} * (0.5 \quad 0 \quad -0.5) = \frac{1}{6} \begin{pmatrix} 1 & 0 & -1 \\ 1 & 0 & -1 \\ 1 & 0 & -1 \end{pmatrix}$$

- y -derivative kernel, H_y^P defined similarly

- Similar to Prewitt, but averaging kernel is higher in middle

$$H_x^S = \frac{1}{4} \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} * \begin{pmatrix} 0.5 & 0 & -0.5 \end{pmatrix} = \frac{1}{8} \begin{pmatrix} 1 & 0 & -1 \\ 2 & 0 & -2 \\ 1 & 0 & -1 \end{pmatrix}$$

$$H_y^S = \frac{1}{4} \begin{pmatrix} 1 & 2 & 1 \end{pmatrix} * \begin{pmatrix} 0.5 \\ 0 \\ -0.5 \end{pmatrix} = \frac{1}{8} \begin{pmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{pmatrix}$$

Prewitt and Sobel Edge Operators

■ Prewitt operator

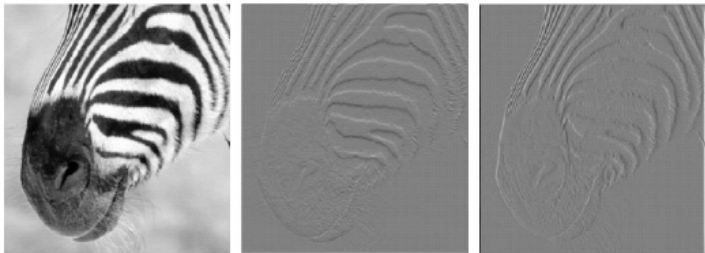
$$H_x^P = \begin{pmatrix} -1 & 0 & 1 \\ -1 & \mathbf{0} & 1 \\ -1 & 0 & 1 \end{pmatrix} \text{ and } H_y^P = \begin{pmatrix} -1 & -1 & -1 \\ 0 & \mathbf{0} & 0 \\ 1 & 1 & 1 \end{pmatrix}$$

$$H_x^P = \begin{pmatrix} 1 \\ \mathbf{1} \\ 1 \end{pmatrix} * \begin{pmatrix} -1 & \mathbf{0} & 1 \end{pmatrix} \text{ and } H_y^P = \begin{pmatrix} 1 & \mathbf{1} & 1 \end{pmatrix} * \begin{pmatrix} -1 \\ \mathbf{0} \\ 1 \end{pmatrix}$$

■ Sobel Operator

$$H_x^S = \begin{pmatrix} -1 & 0 & 1 \\ -2 & \mathbf{0} & 2 \\ -1 & 0 & 1 \end{pmatrix} \text{ and } H_y^S = \begin{pmatrix} -1 & -2 & -1 \\ 0 & \mathbf{0} & 0 \\ 1 & 2 & 1 \end{pmatrix}$$

Prewitt and Sobel Edge Operators



Scaling Edge Components

- Estimates of local gradient components obtained from filter results by appropriate scaling
 - Scaling factor for Prewitt operator

$$\Delta I(u, v) \approx \frac{1}{6} \begin{pmatrix} (I * H_x^P)(u, v) \\ (I * H_y^P)(u, v) \end{pmatrix}$$

- Scaling factor for Sobel operator

$$\Delta I(u, v) \approx \frac{1}{8} \begin{pmatrix} (I * H_x^S)(u, v) \\ (I * H_y^S)(u, v) \end{pmatrix}$$

Gradient-Based Edge Detection

- Compute image derivatives by convolution

$$D_x(u, v) = H_x * I \text{ and } D_y(u, v) = H_y * I$$

- Compute edge gradient magnitude

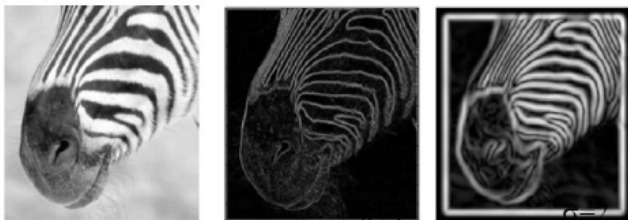
$$E(u, v) = \sqrt{(D_x(u, v))^2 + (D_y(u, v))^2}$$

- Compute edge gradient direction

$$\Phi(u, v) = \tan^{-1}\left(\frac{D_y(u, v)}{D_x(u, v)}\right) = \arctan(D_x(u, v), D_y(u, v))$$

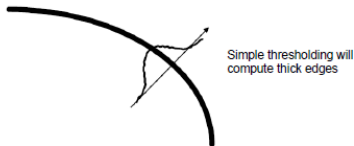
Gradient-Based Edge Detection

- Computing gradient magnitude and orientation
- Mark points where gradient magnitude is large wrt neighbors

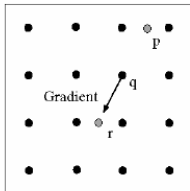


Non-Maxima Suppression

- Retain a point as an edge point if:
 - Its gradient magnitude is higher than a threshold
 - Its gradient magnitude is a local maxima in gradient direction

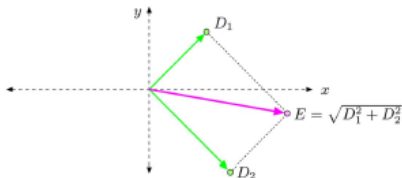


- A maxima occurs at q , if its magnitude is larger than those at p and r



Robert Edge Operator

- Estimates directional gradient along 2 image diagonals
- Edge strength $E(u, v)$: length of vector obtained by adding 2 orthogonal gradient components $D_1(u, v)$ and $D_2(u, v)$



- Filters for edge components

$$H_1^R = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \text{ and } H_2^R = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$$

- Diagonal gradient components produced by 2 Robert filters



$$D_1 = I * H_1^R$$



$$D_2 = I * H_2^R$$

- Digital Image Processing (CS/ECE 545), Computer Science Dept. Worcester Polytechnic Institute (WPI)